

RENU VELHAL

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EDUCATION

California State University, Fullerton

Aug 2022 – May 2024

Master of Science in Computer Science

Courses: Advanced Algorithms, Artificial Intelligence, Advanced Database Management, Modern Software Management

EXPERIENCE

Software Engineer, The Sunrise State | Los Angeles, California

August 2024 – Present

- Built the backend flow for a checkout system using 10+ RESTful APIs with Node.js and Express to handle payment processing and order management. Implemented caching and load balancing, reducing response times by 40% and scaling to support daily requests
- Transitioned critical API endpoints from REST to GraphQL to enable flexible data querying and eliminate redundant calls. Improved efficiency by reducing API requests by 50%, increasing engagement by 25%, and expediting updates by 35%
- Developed a library of 15+ reusable React components utilizing hooks and Redux to tackle inconsistent frontend structures. Unified workflows reduced duplication, cutting development effort by 30% and accelerating feature deployment timelines
- Enhanced MongoDB performance by redesigning schemas and introducing advanced indexing techniques to manage high transactional volumes. Achieved 30% faster query execution, ensuring seamless data handling during real-time updates
- Incorporated Kubernetes for scalable resource management and Jenkins for CI/CD pipelines to handle peak demand seamlessly. Automated deployments reduced system downtime to zero and cut release cycles from 3 days to 4 hours

Software Engineer, Capgemini | Pune, India

September 2021 – August 2022

- Led the development of a real-time messaging system for McDonald's using Socket.IO and Redis. Optimized history loading and connection pooling, achieving 30% faster message delivery and ensuring timely order communication during high-demand periods
- Designed and implemented over 20 React components to resolve fragmented user interface designs. Applied React Context for state management of cart and order statuses, achieving real-time synchronization across pages and improving user experience
- Created the useOrderStatus custom hook for managing live order updates through WebSocket connections. Achieved synchronization of order data while minimizing redundant re-fetches, streamlining the logic by 40% and providing accurate real-time tracking
- Optimized backend API performance with Express.js and MongoDB to address bottlenecks in retrieving order and inventory data. Introduced advanced schema design and indexing strategies, resulting in consistent query response times of 50ms under high traffic

Software Engineer, Technocolabs | Kolhapur, India

September 2019 – August 2021

- Devised responsive, SEO-friendly web applications using React and Node.js to resolve slow page loads. Refined rendering logic and API efficiency, reducing load times by 60% and increasing user retention by 40% for growth
- Developed a robust JWT-based authentication framework with rate-limiting to secure access for critical order management systems. Prevented unauthorized access by 95% through carefully architected token validation workflows and advanced access control policies
- Streamlined deployment pipelines using Jenkins and Docker on GCP to accelerate releases and minimize errors. Automated testing and integration processes, reducing deployment time by 40% while maintaining high performance and application consistency
- Implemented comprehensive unit tests with Jest and React Testing Library to ensure reliable component functionality. Crafted edge-case test scenarios, achieving 90% coverage and reducing production issues by 30%, improving application quality and stability
- Conducted stress tests and load simulations on ordering systems during peak traffic periods such as promotions. Identified and resolved critical performance bottlenecks, scaling the platform to handle 2 million daily requests without service disruptions

PROJECT

Playlist Keyword Scan

February 2024 – May 2024

- Innovated a Chrome extension with React.js and Node.js to filter playlist content based on keywords. Enhanced search functionality by 40% and added persistent search history, improving user experience and boosting session efficiency
- Configured Node.js APIs using Express to handle search history management and retrieval for seamless session continuity. Integrated React.js for real-time keyword filtering, delivering a smooth user interface and improving usability significantly

TECHNICAL SKILLS

Languages: Java, C++, JavaScript, Typescript, Python, SQL, Golang

Web development: HTML, CSS, React.js, Node.js, Express, Django, Next.js

Databases: MongoDB, PostgreSQL, Redis

Technologies: Git Version Control, Redux, AWS, CI/CD, Docker, Google Cloud